

Tran Quoc Truong

Product Engineer · Applied AI · Backend Systems

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Summary

Product Engineer & Applied AI Engineer translating real business and operational problems into reliable AI systems. Designs end-to-end solutions across agentic workflows, retrieval, model adaptation, backend architecture, and evaluation, with hands-on ownership from system design through delivery.

Technical Skills

AI & Agentic Systems: RAG, Hybrid Retrieval, LangGraph, MCP (Model Context Protocol), Tool Calling, LiteLLM, Qdrant, Docling
ML, Fine-Tuning & Model Serving: PyTorch, Hugging Face Transformers, PEFT, Unsloth, LoRA/QLoRA, vLLM, Synthetic Data Generation
LLM Evaluation & Observability: OpenTelemetry (OTel), Runtime Trace Reconstruction, LLM-as-a-Judge, Regression Testing, Deterministic Quality Gates, Langfuse
Backend Engineering: Python, FastAPI, Django/DRF, Pydantic, SQLAlchemy, PostgreSQL, Redis, Celery, SSE / Streaming APIs
Infrastructure & Delivery: Docker, Docker Compose, RunPod, Nginx, MinIO, CI/CD, Linux

Experience

FPT Software , Applied AI Engineer & Automation Tester	Ho Chi Minh City, Vietnam Dec 2024 – present 1 year 11 months
IQP (Code Intelligence & Quality Retrieval): Designed multi-repository code intelligence combining hybrid retrieval, Code Graphs, and MCP to provide traceable engineering context for AI-assisted quality analysis. Agent Assurance: Translated AI reliability requirements into an assurance engine for trace-based behavioral evaluation, reproducible regression, and governed quality gates. Banking Automation & Quality Operations: Engineered reusable cross-platform test automation and 1,500+ lines of SQL validation across banking and payment systems, cutting manual regression effort by ~70%.	
FPT Software Academy , Full-Stack Developer Intern	Ho Chi Minh City, Vietnam Sept 2024 – Dec 2024 4 months
- Built the core backend for an online learning and assessment platform using Django with role-based access control and automated exam workflows. - Integrated a separate FastAPI computer-vision service for AI-assisted real-time exam proctoring and face detection. - Containerized and deployed the multi-service architecture using Docker Compose, Nginx reverse proxy, and Cloudflare Tunnel.	

Achievements

Golden Solution Prize (1st Place) — IVS Solution Day 2.0 2026 Golden Solution Prize for Flezi Polaris, an Agent Assurance solution for trace-based AI evaluation and governed quality verification.	Sept 2026
2nd Place — IVS Hackathon 2026 2nd place for independently architecting the Agent Assurance engine later integrated into Omni-Agent.	Aug 2026
Second Prize — Autonomous Driving Research Paper Competition Second Prize for sole-authored autonomous-driving research combining YOLOv8 perception, lane segmentation, and PID control.	Apr 2024

Featured Projects

Omni-Agent — Applied AI Platform (Knowledge Builder & Agent Assurance)

An Applied AI platform engineered from real operational needs, unifying versioned document RAG, LangGraph multi-agent orchestration, and integrated agent verification within a modular multi-service architecture.

- Platform Architecture: Engineered a modular monorepo featuring a Django/DRF control plane for project tenancy, identity, and permissions, backed by decoupled, stateless Celery workers for asynchronous processing workloads.
- Knowledge Builder: Built document processing pipeline with Docling layout parsing/OCR, snapshot chunking, Qdrant vector indexing, a Streamable HTTP MCP tool server, and a LangGraph-based multi-agent runtime with citation-grounded RAG.
- Agent Assurance Integration: Integrated execution evidence, regression evaluation, and governed quality gates as a native platform capability.
- Reliability & Data Integrity: Designed idempotent evidence ingestion keyed by trace/span identity, with PostgreSQL-backed execution metadata and S3-compatible artifact storage.

APIT — Agent Programmatic Integration Testing

AI-assisted API testing framework generating structured test scenarios from API documentation via fine-tuned LLMs.

- Developed a bilingual synthetic dataset generation pipeline producing 1,258 domain-specific API testing samples.
- Fine-tuned Qwen2.5-3B with LoRA/QLoRA, achieving macro-F1 of 0.655 (+41.5% relative improvement over Llama-3.2-3B baseline).
- Deployed the fine-tuned model on RunPod through an OpenAI-compatible vLLM service with quantized inference and runtime LoRA adapters.
- Implemented a hybrid evaluation pipeline combining LLM-as-a-Judge, fuzzy matching, and deterministic JSON-schema validation.

E-Commerce AI Assistant — Conversational RAG & Search

End-to-end shopping assistant combining filtered vector search and multi-agent conversational retrieval.

- Built product semantic search using Milvus, cosine similarity, embeddings, and dynamic category/price filters.
- Implemented multi-turn conversational agent with LangGraph and LangChain for context-aware product consultation.
- Containerized multi-service architecture with FastAPI, PostgreSQL, Redis, Milvus, and Docker Compose.

Education

FPT School of Business & Technology, Master of Software Engineering in AI	Ho Chi Minh City, Vietnam May 2026 – present
FPT University, Bachelor of Artificial Intelligence	Ho Chi Minh City, Vietnam 2021 – 2025

Languages

- Vietnamese:** Native
- English:** Professional working proficiency